Status	Finished
Started	Sunday, 25 October 2026, 6:42 PM
Completed	Sunday, 25 October 2026, 6:45 PM
Duration	3 mins 13 secs
Grade	**13.00** out of 15.00 (**86.67**%)

Question **1**

Complete

Mark 1.00 out of 1.00

What is the primary advantage of the divide-and-conquer approach in sorting algorithms?

a. It simplifies the sorting process by using only one pass
b. It allows for efficient parallel processing and sorting of data
c. It avoids the need for recursion
d. It only works on small datasets

The correct answer is: It allows for efficient parallel processing and sorting of data

Question **2**

Complete

Mark 1.00 out of 1.00

What is one advantage of sorting a list before performing a search operation?

- a. It allows for faster searching
- b. It increases the number of comparisons needed
- c. It has no effect on the search operation
- d. It makes the search operation slower

The correct answer is: It allows for faster searching

Question **3**

Complete

Mark 0.00 out of 1.00

Two-way merge sort algorithm is used to sort the following elements in ascending order.
200,470,150,80,90,40,400,300,120,70
What is the order of these elements after second pass of the merge sort algorithm?

- a. 40,70,80,90,120,150,200,300,400,470
- b. 200,470,80,150,40,90,300,400,70,120
- c. 40,80,90,150,200,300,400,470,70,120
- d. 80,150,200,470,40,90,300,400,70,120

The correct answer is:
80,150,200,470,40,90,300,400,70,120

Question **4**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the process of Merge Sort?

- a. It divides the list into two halves, sorts each half, and then merges them
- b. It compares adjacent elements and swaps them if necessary
- c. It repeatedly finds the minimum element and moves it to the sorted part of the list
- d. It builds a sorted array one element at a time

The correct answer is: It divides the list into two halves, sorts each half, and then merges them

Question **5**

Complete

Mark 1.00 out of 1.00

_____is putting an element in the appropriate place in a sorted list yields a larger sorted order list.

- a. Insertion
- b. Extraction
- c. Selection
- d. Distribution

The correct answer is:
Insertion

Question **6**

Complete

Mark 1.00 out of 1.00

Which of the following is a key reason for the importance of sorting algorithms?

a. Sorting decreases the efficiency of selection operations
b. Sorting is rarely used in programming
c. Sorting helps in finding duplicates quickly
d. Sorting makes it harder to search for items

The correct answer is: Sorting helps in finding duplicates quickly

Question **7**

Complete

Mark 1.00 out of 1.00

What is one of the key advantages of using the built-in sorted() function in Python?

a. It only works with integer arrays
b. It is less efficient than custom sorting algorithms
c. It requires external libraries
d. It sorts data out of the box efficiently

The correct answer is: It sorts data out of the box efficiently

Question **8**

Complete

Mark 1.00 out of 1.00

What is sorting in the context of computer science?

- a. Arranging data in a particular format
- b. Searching for data in a list
- c. Deleting data from a list
- d. Inserting data into a list

The correct answer is: Arranging data in a particular format

Question **9**

Complete

Mark 1.00 out of 1.00

Algorithm design technique used in merge sort algorithm is

- a. Dynamic programming
- b. Greedy method
- c. Divide and conquer
- d. Backtracking

The correct answer is:
Divide and conquer

Question **10**

Complete

Mark 1.00 out of 1.00

Which Python function would you use to sort a list in-place?

- a. arrange()
- b. sort()
- c. order()
- d. sorted()

The correct answer is: sort()

Question **11**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after the two halves of the list are sorted?

- a. They are compared element by element
- b. They are combined to form a single sorted list
- c. They are discarded
- d. They are split again into smaller sublists

The correct answer is: They are combined to form a single sorted list

Question **12**

Complete

Mark 0.00 out of 1.00

Very slow way of sorting is_____

- a. Bubble sort
- b. Quick sort
- c. Heap sort
- d. Insertion sort

The correct answer is:
Insertion sort

Question **13**

Complete

Mark 1.00 out of 1.00

The process of placing or rearranging a collection of elements into a particular order is known as

- a. Rearranging
- b. Merging
- c. Searching
- d. Sorting

The correct answer is: Sorting

Question **14**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is based on repeatedly dividing the list into halves?

a. Quick Sort
b. Insertion Sort
c. Merge Sort
d. Bubble Sort

The correct answer is: Merge Sort

Question **15**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm involves comparing elements and swapping adjacent items that are out of order?

a. Merge Sort
b. Binary Search
c. Linear Search
d. Bubble Sort

The correct answer is: Bubble Sort